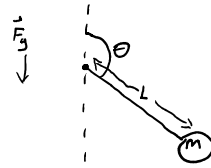


2.1.5 a)

The equation of motion for a damped pendulum confined to a plane is

$$mL\ddot{\theta} = mg \sin[\theta] - bL\dot{\theta}$$

where  $b$  is the strength of viscous drag.



This eq. can be nondimensionalized by rescaling time with  $\tilde{t} = tT$ , we get:

$$T^2 \frac{L}{g} \frac{d^2 \theta}{d\tilde{t}^2} = \sin[\theta] - \frac{bL}{mg} T \frac{d\theta}{d\tilde{t}}$$

If we set our scaling to be  $T = \frac{mg}{bL}$ , this eq. becomes

$$\frac{m^2 g}{b^2 L} \ddot{\theta} = \sin[\theta] - \dot{\theta}.$$

If  $\frac{m^2 g}{b^2 L} \ll 1$ , then the  $\ddot{\theta}$  term can be ignored and we have

$$\dot{\theta} = \sin[\theta].$$

This means that a pendulum whose drag is much larger than its inertia can be considered to be described by the 1<sup>st</sup> order ODE in the problem where  $x = \theta$ .

b) In light of this physical analogue, it is clear that  $x=0$  is an inverted pendulum and therefore unstable. Similarly,  $x=\pi$  is at the bottom and stable.

2.2.10

a)  $f(x) = 0$

b)  $f(x) = a \sin[\pi x]$ , where  $a$  is a real, non-zero number

c) This is not possible. Fixed points correspond to points where  $f$  changes signs and the stability depends on which way it changes. Since  $f(x)$  must be a smooth function of  $x$ , there is no way to construct a function that crosses zero from the same direction consecutively (in 1-b).

d)  $f(x) = c$  where  $c$  is a non-zero constant.

e)  $f(x) = (x - c_1)(x - c_2)(x - c_3) \dots (x - c_{100})$

2.3.3

$$a) \quad \frac{dN}{dt} = -a N \ln[bN] \rightarrow -\frac{1}{a} \int dN \frac{1}{N \ln[bN]} = \int dt$$

$$\Rightarrow -\frac{1}{a} \ln[\ln[bN]] = t + c$$

$$\ln[bN] = e^{-at+c}$$

$$\boxed{N = \frac{1}{b} e^{-at+c}}$$

So,  $a$  sets the growth rate

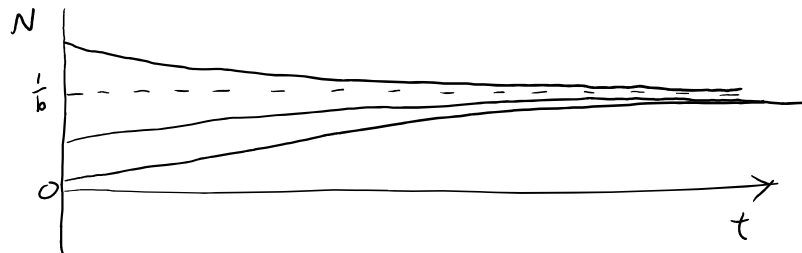
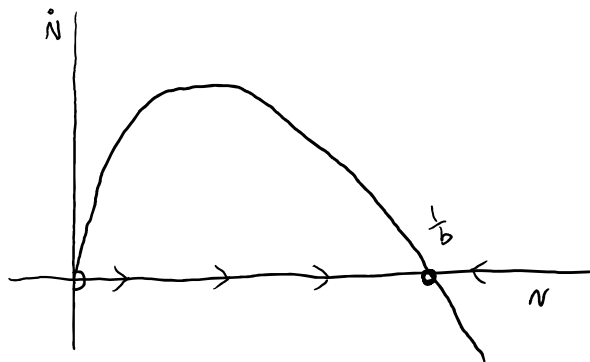
Similarly,  $\dot{N} = 0 \Rightarrow 0 = -a N (\ln[b] + \ln[N])$

$$\Rightarrow N = 0 \quad \text{and} \quad \ln[b] + \ln[N] = 0$$

$$\hookrightarrow N = \frac{1}{b}$$

So,  $\frac{1}{b}$  sets the stable tumor size.

b)



2.4.2

$$\dot{x} = x(1-x)(2-x)$$

$$f(x^*) = 0 \Rightarrow x^* = 0, 1, 2$$

$$f'(x) = 3x^2 - 6x + 2$$

$$\Rightarrow f'(x^*=0) = 2, \text{ unstable}$$

$$f'(x^*=1) = -1, \text{ stable}$$

$$f'(x^*=2) = 2, \text{ unstable}$$

**2.4.9** a)

$$\frac{dX}{dt} = -X^3$$

$$\Rightarrow \int dx \frac{1}{x^3} = \int dt$$

$$\frac{1}{2x^2} = t + C$$

$$\Rightarrow x = \frac{\pm 1}{\sqrt{2(t+C)}}$$

$$\lim_{t \rightarrow \infty} x(t) = \frac{\pm 1}{\sqrt{2t}} = 0$$

b)

